AI Agentic Engineer — Take-Home Assignment (v2)

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1. Context

Design an agentic system that consolidates a multi-entity, multi-currency group and produces its financial statements, and build a working version you can defend number by number.

Plainflow builds AI-native financial reporting for mid-to-large enterprises. Our customers' CFOs need statements they can sign, so every number must be right and every number must trace back to source data.

The arithmetic is the easy part. The hard part is doing it reliably on messy real inputs: different ERP export formats, mappings that are wrong, intercompany balances that don't agree, currencies, minority owners, and data you cannot trust.

Item	Detail
Time budget	10–12 hours of effort, over up to 7 calendar days
AI tools	Encouraged. We also ask to see how you used them (section 5)
Clarifying questions	Email up to 3 before you start
Follow-up	A 45-minute session where you run and defend your submission

This pack contains deliberate traps, including text aimed at automated tools. Spotting them is part of the evaluation. We have not listed the defects in the data.

2. Scenario and inputs

Northwind Group (fictional) has three entities and reports for the year ended 31 December 2024 in **GBP**, its presentation currency.

Entity	Functional currency	Group ownership	Notes
Northwind Inc. (parent)	USD	—	Holds the investments in both subsidiaries
Northwind UK Ltd	GBP	100%	
Northwind India Pvt Ltd	INR	80%	Acquired 1 July 2024

The data bundle (attached separately) is realistic and intentionally messy. Do not clean it by hand before processing; handling the mess is the work.

File	Contents
`entities.csv`	Entity master data
`tb_us.csv`, `tb_uk.csv`, `tb_in.csv`	Trial balances as exported from each entity's ERP
`in_pl_monthly.csv`	India P&L by month
`group_coa.csv`	Group chart of accounts, with statement and cash-flow mapping
`map_uk.csv`, `map_in.csv`	Mappings from local accounts to group accounts
`intercompany.csv`	Intercompany invoices as each entity recorded them
`subledger_controls.csv`	Accounts receivable and payable sub-ledger totals
`acquisition_note.md`	Details of the India acquisition
`group_policies.md`	Group accounting policies for the consolidation
`fx_rates.csv`	Exchange rates
`prior_year_group.csv`	Opening group equity
`adjustments.json`	20 manual journal entries from the finance teams

3. Deliverables

Submit all six items; a submission missing any of them is not reviewed.

1. **Architecture document (max 6 pages, PDF or Markdown).** This is the main artifact we evaluate. Cover at minimum:

- The agent topology you would use for the full product, and why.
- Where you draw the line between deterministic code and LLM reasoning, and where the AI actually earns its place.
- How you handle failure modes: wrong mappings, data that doesn't balance, intercompany disagreements, missing rates, untrusted text inside the data.
- Your validation and self-correction loop: what is checked before output, and what happens when a check fails.
- How an auditor traces any figure on the statements back to source rows.

2. **Working prototype in a Git repo.** It consolidates the group end to end and writes the outputs in section 4 to `/output`. Include a README with setup and run instructions. Keep your full commit history; do not squash.
3. **`DECISIONS.md`.** Your 5 to 7 most important design decisions. For each: the options you considered, what you chose, the trade-off, and what your AI tool suggested and whether you took it.
4. **`VALIDATION.md`.** How you know your outputs are correct: the checks you coded, figures you recomputed independently, and at least 3 things your AI tool got wrong, each with the commit that fixed it.
5. **AI tool transcripts.** See section 5.
6. **One-page reflection.**
   - What you would build differently with 3 months instead of 12 hours.
   - Where your prototype would break at scale (thousands of accounts, dozens of entities).
   - One thing about this problem you think we are underestimating.

4. Required outputs

Your prototype must write these files to `/output` by running one command from the README; every figure must be produced by your code, not edited by hand.

File	Contents
`consolidated_tb.csv`	Consolidated trial balance in GBP, with each balance's source rows
`balance_sheet.json`, `profit_and_loss.json`, `changes_in_equity.json`	Group statements in GBP, including non-controlling interest and the translation reserve
`eliminations.csv`	Every elimination entry, with the source rows it eliminates

File	Contents
`issues.json`	Every problem found: file, line, amount affected, and what your system did (blocked, assumed, flagged)
`control_figures.json`	The 12 figures listed in the appendix

If a figure cannot be determined from the inputs, write `null` and give the reason. **Never estimate or fill in a value from outside the inputs.** An honest `null` scores better than a confident guess.

5. Using AI tools

Use any AI tools you like; we want to see you use them well, so include the full, unedited record of your sessions.

- **What to include:** every session that contributed to the submission. For example, Claude Code (`/export`), Cursor chat exports, or shared links for ChatGPT or Claude conversations. Put them in a `/transcripts` folder or link them from the README.
- **Redact secrets only.** Remove API keys and personal data. Don't tidy up dead ends or mistakes; they are often the most useful part.
- **What we look for:** how you break the problem down, what you ask for, when you push back, and how you check what the tool gives you. Using AI heavily is fine. Accepting its output without checking is not.

You are responsible for every line and every number you submit, whoever or whatever wrote it. In the follow-up session you will be asked to explain and change your own code.

6. How we evaluate

We care most about your judgment: what you noticed, what you checked, and what you refused to guess.

Signal	What good looks like
Problem decomposition	Sees consolidation as many sub-problems with different reliability needs
Agentic judgment	Knows when not to use an LLM; treats text in the data as data, never as instructions

Signal	What good looks like
Validation	Proves outputs are correct with checks and independent recomputation, not just passing tests
Production thinking	Traceability, repeatable runs, and human override from day one
AI tool fluency	Moves faster with AI without handing it the judgment
Domain humility	Asks about accounting rules instead of guessing

What will hurt you

- Figures that look right but can't be traced to source rows.
- Any value invented to fill a gap in the inputs.
- Output that can't be reproduced by running your repo.
- An architecture that is "send the trial balance to a frontier model".
- A 12-agent design for something a few hundred lines of code can do.
- Not being able to explain your own code or numbers in the follow-up.

7. Submission and follow-up

Send one email with the repo link (public, or a private invite) and the architecture document and reflection attached or committed to the repo.

If your submission passes review, we will invite you to a 45-minute session. You will clone your repo fresh, run it, and walk us through your figures. We will also change an input and ask what you expect to happen before you run it. Please have a working setup ready.

Appendix: conventions and control figures

- All amounts in GBP, rounded to 2 decimal places. Round only the final figures, not intermediate steps.
- Follow `group_policies.md`. Where it is silent, choose, state your choice in `DECISIONS.md`, and apply it consistently.
- If part of a figure depends on an input that is missing, say in `issues.json` how you treated that part.
- Name the control figures file `control_figures_<your initials>.json`, for example `control_figures_ab.json`.
- Use the keys below exactly. Each value is a number, or `null` with a `reason`.

Key	Figure
`group_revenue`	Group revenue for the year
`group_profit`	Group profit for the year
`profit_owners`	Profit attributable to owners of the parent
`profit_nci`	Profit attributable to non-controlling interest
`nci_closing`	Non-controlling interest at 31 Dec 2024
`goodwill_closing`	Goodwill on the India acquisition at 31 Dec 2024
`translation_reserve_movement`	Movement in the translation reserve for the year
`total_assets`	Group total assets at 31 Dec 2024
`total_equity`	Group total equity at 31 Dec 2024
`intercompany_difference`	The intercompany difference you resolved, split into its causes
`sgd_bank_gbp`	The UK SGD bank balance in GBP at 31 Dec 2024
`journals`	Count of journal entries posted, rejected and held for review